

Physiological Recording (Spike)

About



The FIL has provisions for recording basic physiological data (heart rate & breathing) within the MRI, MEG and Testing Room Environments.

In all environments a 1401 micro-controller and Spike v2 software from [Cambridge Electronic Solutions \(CED\)](#) facilitate recording.

All MRI and the MEG units have their own physiological recording system installed and ready for use. One physiological recording system is kept on a portable trolley and can be booked for use in FIL Testing Rooms.

The CED 1401 micro-controller and Spike 2 software is versatile and can be adapted to record eye-tracking, air-pressures, voltages from other peripheral equipment etc. Such use is non-standard and requires approval from the Cognitive Interface Team.

User Guide

1. Order of Events for Physiological Recording

It is important that the recording of imaging data and physiological data are initiated in the correct order:

[follow these instructions](#)

Preparation

Prior to entry: Ensure all persons are completely metal free. Belongings should be in lockers.

1. Inform Imaging Support staff you wish to record physiological data
2. Imaging Support staff will perform safety screening of the volunteer
3. Imaging support Staff will position the volunteer for imaging, applying the breathing belt and pulse oximeter

Data Collection Steps

1. Start Spike2 recording
2. Start stimulus script
3. Instruct scan to start
4. Ensure script triggers correctly
5. Ensure physio data is recording correctly
6. Save Spike2 recording as .smr file

7. Repeat steps 1 to 6 until all data is collected

Procedure End

1. Volunteer will be removed from the imaging suite .
2. Save all Spike data to a personal drive CHER (\\cher\<your username>).
3. Shutdown the Spike PC

2. Spike v2 Operation

When the correct order of events is understood, continue to operate the Spike2 software to make recordings:

[follow these instructions](#)

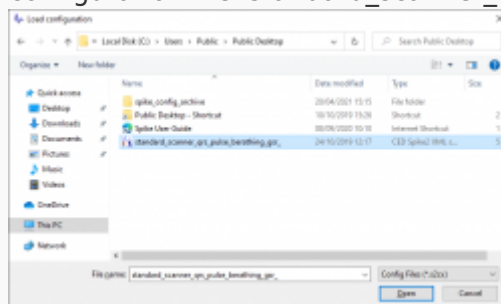
1. Open Spike Software:

1. Login to Spike2 PC (with FIL Credentials)
2. From the Desktop double click 'Spike2' to open Spike software

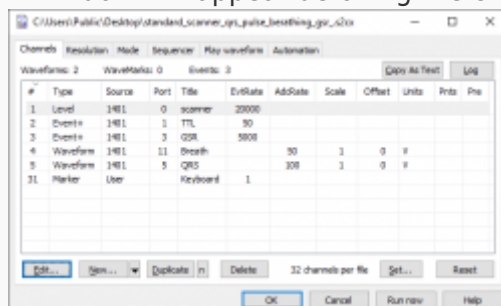


2. Set the Configuration File:

1. A configuration file sets what physiological data to record and how to record it
2. A standard configuration file can be found on the desktop of each Spike Laptop
3. File > Load Configuration > Navigate to 'C:\Desktop\Public Desktop' > Choose configuration file 'standard_scanner_gsr_pulse_breathing_gsr' > Click 'Open'



4. A window will appear detailing the channels used by the configuration > Click 'OK'

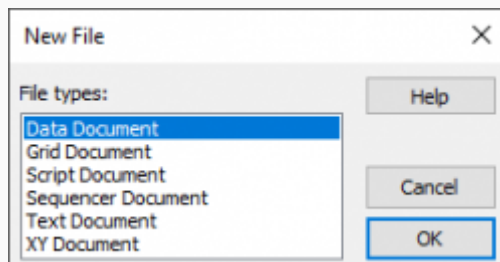


5. The configuration file is now set.
6. This will need to be loaded each time the Spike software is opened.
7. **Note:** old or custom configuration files can be located here 'C:\Desktop\Public Desktop\spike_config_archive', These should only be used if instructed by cognitive

interface.

3. Open a Data Document:

1. Open a new Data Document, this is a blank environment in which to store data.
2. File > New > Data Document

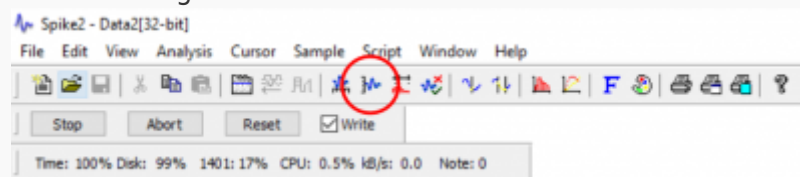


4. Record Data:

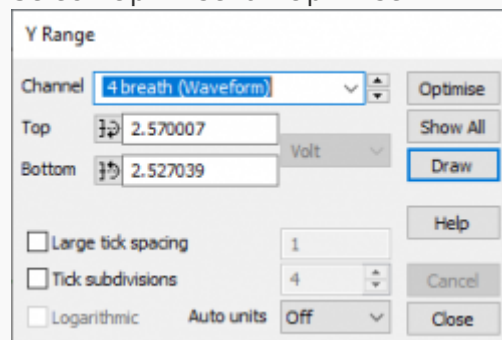
1. Before starting the MRI scan
 1. Prepare any stimulus script (on cogent etc)
 2. On the Spike software > Click 'Start' to begin recording



3. At this point check to ensure data is recording as expected.
4. You may need to optimise the Y-Axis to best visualise data.
 1. Click 'Y-Range' icon



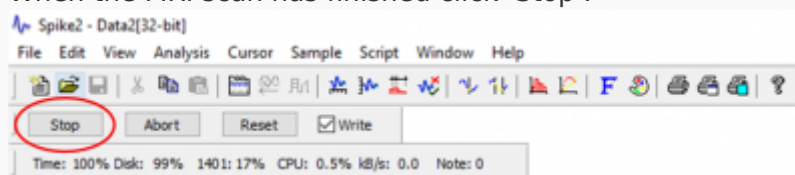
2. Select 'Optimise' or 'Optimise All'



5. Now instruct Imaging Support to start the MRI sequence.

5. Save Data:

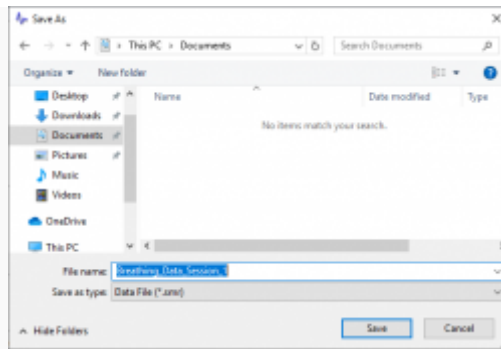
1. When the MRI scan has finished click 'Stop'.



2. Click the 'Save' icon.



3. Select the directory you wish to use
4. Name the file as desired
5. Click 'Save'



6. Check Data:

1. Check the save directory and you will see two files:
2. **.smr** = large file containing the physiological data
3. **.s2r** = smaller file containing the screen format and display settings
4. Note: the .s2r file is generated when a data document is closed from within Spike or when closing the Spike software.

7. Start a New Recording:

1. To start a new recording using the previously loaded configuration file
2. File > New > Data Document

8. Save Data to External Source:

1. Save all Spike data files to a personal drive.

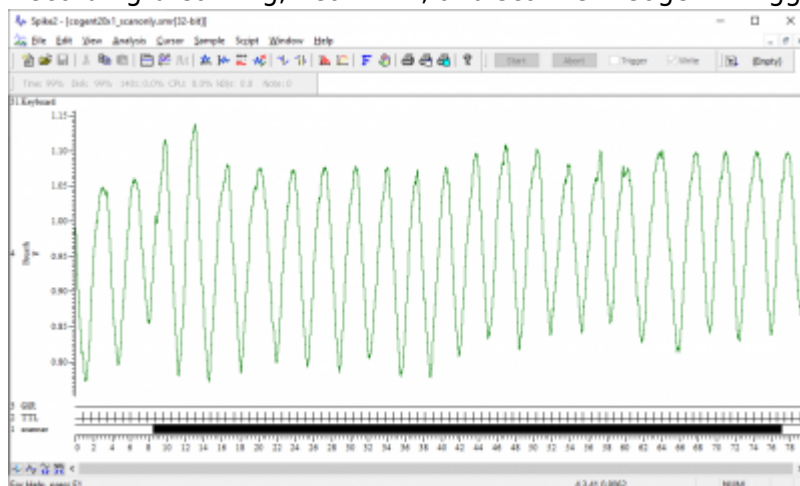
Manufacturer Resources

Spike2 software tutorials are available on the [CED tutorials page](#).
Spike2 software manuals are available on the [CED downloads page](#).

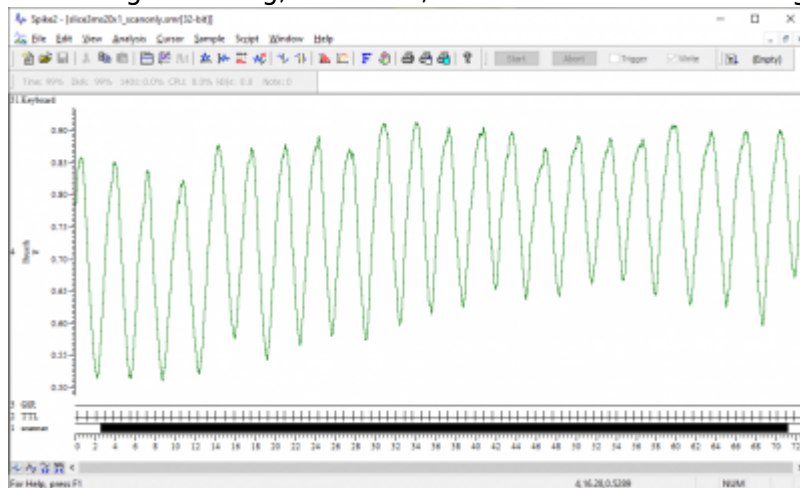
Spike Trace Examples

Example Screenshots

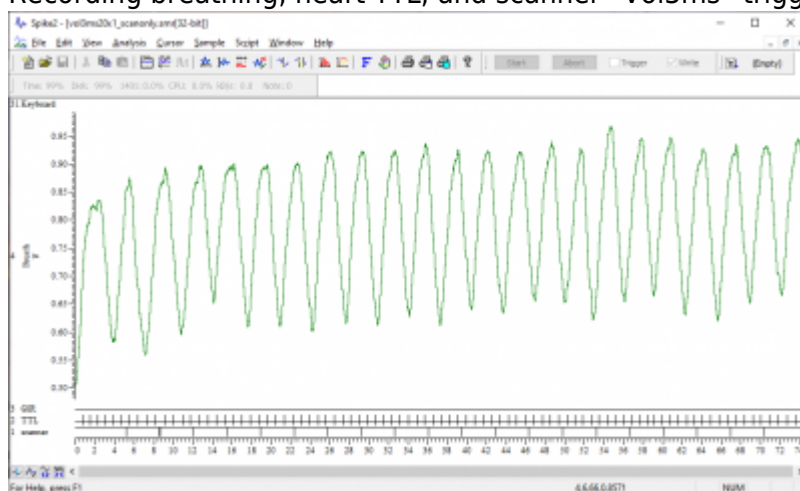
Recording breathing, heart TTL, and scanner "Cogent" triggers



Recording breathing, heart TTL, and scanner "Slice3ms" triggers



Recording breathing, heart TTL, and scanner "Vol3ms" triggers



Spike Configuration Files

Prior to recording via the Spike software a 'configuration file' must be selected. This will configure which 1401 channels are set to record and how.

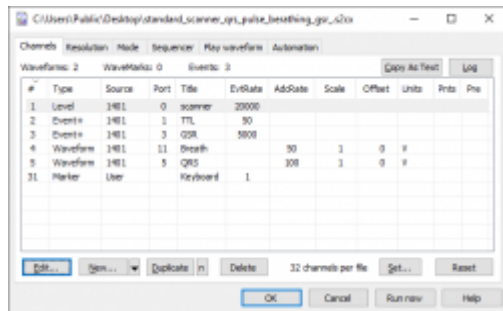
In an effort to simplify operations, we have moved to using a single configuration file:

File Name: standard_scanner_qrs_pulse_breathing_gsr

File directory: C:\Desktop\Public Desktop

This single configuration will record the following data:

- Scanner Triggers (TTL)
- Cardiac pulse (TTL)
- GSR (waveform)
- Breathing (waveform)
- Cardiac pulse (QRS waveform)



On request a non-standard configuration file can be generated for individual study use, these may include:

- Audio
- Eye tracking
- Transducers

Spike Data Files

SPM's Physiological Noise Correction Toolbox requires Spike data to be saved in Spike's native SMR format. This is the default when saving data in Spike.

These files can be read by the Spike software, should a future inspection be required.

Within the chosen save directory two files will be generated:

- **.smr** = large file containing the physiological data
- **.s2r** = smaller file containing the screen format and display settings

Note: the .s2r file is generated when a data document is closed from within Spike or when closing the Spike software.

Upon finishing a recording session move all Spike Data Files to an external location i.e a personal drive or CHER (\\cher\<your username>).

An example Spike data file with scanner pulses, QRS waveform, Breathing waveform, heartbeat TTL, and scanner pulses (55 volumes, "Vol3ms") is available [here](#)

Spike Data File Analysis

Use the SPM Physiological Noise Correction Toolbox (available here: <svn://bread.fil.ion.ucl.ac.uk/tools/physio/trunk>) to read and analyse the Spike data files. This toolbox will create regressors for heartbeat wrt scanner pulses, and breathing wrt scanner pulses. The toolbox is compatible with both "Cogent" scanner triggers and "Slice3ms" scanner triggers and should auto detect which trigger type was in use.

Note: The trigger type, found under sequence / special / trigger type on the scanner console, will be selected and saved for your protocol during your pilot scan.

If your sequence was run with “Vol3ms” triggers then please use Nadege's code to create the regressors. (link TBC)

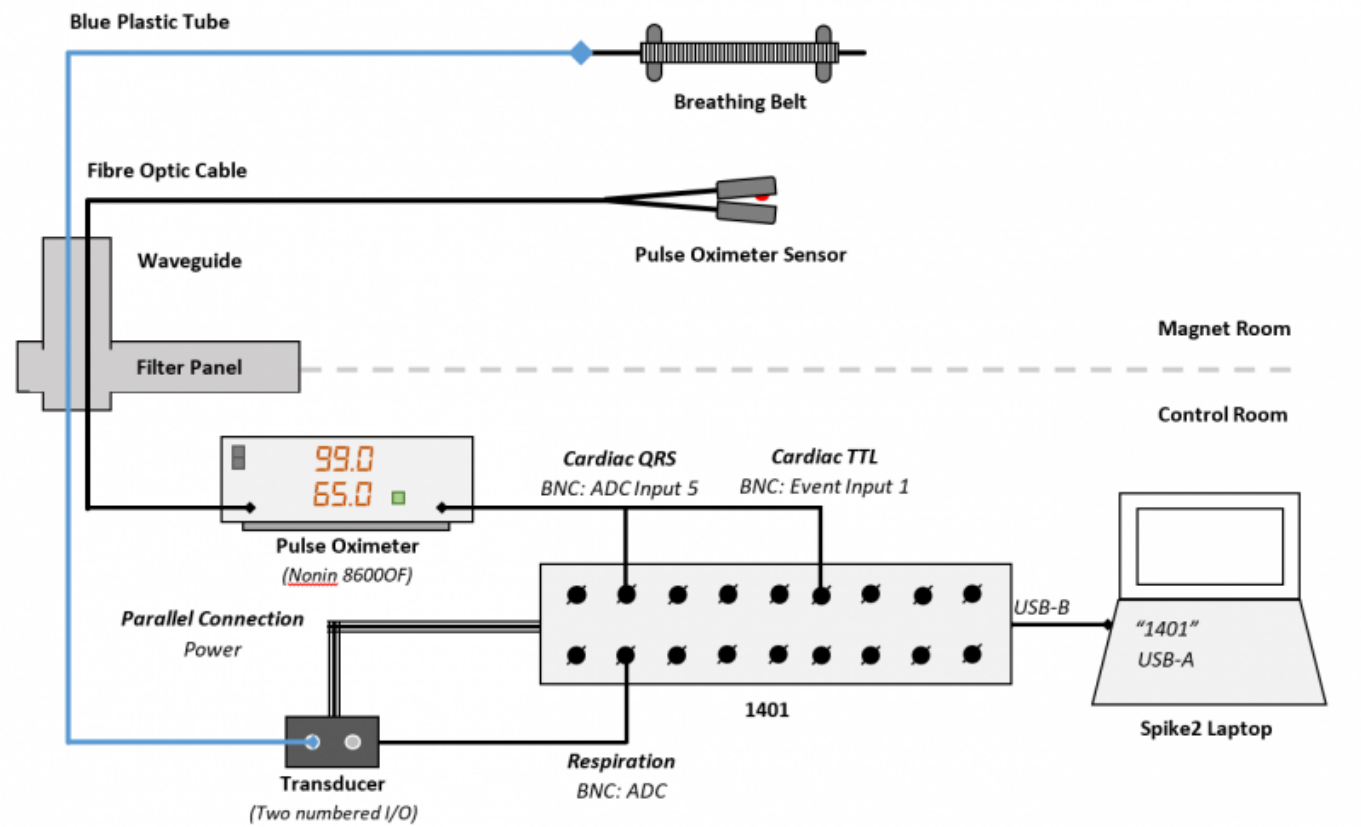
Equipment

Physiological Recording System

Physiological Recording System					
CED Micro1401 (1401)	Spike laptop Running Spike2 software (CED)	Nonin 8600 Pulse Oximeter	Nonin Pulse Oximeter Fibre Sensor	Breathing Belt	Breathing Belt Transducer produced by FIL
					

Configuration

A connection diagram of the basic physiological recording system can be seen below:



Troubleshooting

Spike Laptop Cannot Detect 1401 Device

Quattro Only Check connections:

1. Ensure 1401A is 'ON'
 1. If necessary: connect the 3-pin plug labeled '1401B' into a free wall socket
 2. Green LED will be lit when powered.
2. Ensure 1401B USB is connected to Spike2 laptop
 1. Unplug the USB lead labeled '1401A'
 2. Connect the USB lead labeled '1401B'
3. Close Spike2 software and re-open

If connections are correct, restart Spike2 laptop:

1. Power 'OFF' the Spike2 laptop
 1. Ensure 1401B remains ON and connected to the laptop
 2. **Important:** Logout all other users, open profiles may be connected to a 1401

If error remains, run software 'Find1401':

1. If open, close Spike2 software
2. Run the software 'Find1401'
 1. Click find
 2. Close 'Find1401' software
3. Open Spike2 software

If error remains contact the Cognitive Interface Team.

No Trace is Seen on Spike Software

Check the input scaling on Spike2:

1. Select desired channel
2. From the tool bar click #####
3. Then select 'Optimise'

Check the connections:

1. Ensure breathing belt transducer is properly connected to the blue plastic tube
2. Ensure the breathing belt cable is not impinging by the volunteer of scan bed.
3. Ensure Nonin pulse oximeter is switched 'ON'

Check the Spike2 configuration file:

1. Close and open Spike2 software

2. Load new configuration file
3. Ensure the correct file has been loaded

Poor Pulse Signal

Re-attach pulse oximeter securely:

1. Use micropore tape
2. Ensure red light is facing the nail bed
3. Ensure the velcro cover is used to remove light interference

Other tricks:

1. Place hand under blanket to keep warm and dark
2. Remove nail varnish
3. Try a different finger - choosing one where the sensor light sits over the nailbed
4. Keep the hand still

From:

<https://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/> - **Cognitive Interface Wiki**

Permanent link:

https://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/doku.php?id=physiological_recording

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